PROBABILITIES (CH 6 & 7)

For $X \sim N(\mu, \sigma)$ and $\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

$P(X < a)$	$= NORM.DIST(a, mean, standard\ deviation, TRUE)$
$P(X > a)$ $= 1 - P(X < a)$	$= 1 - NORM.DIST(a, mean, standard\ deviation, TRUE)$
$P(a < X < b)$ $= P(X < b) - P(X < a)$	$= NORM.DIST(b, mean, standard\ deviation, TRUE)$ $- NORM.DIST(a, mean, standard\ deviation, TRUE)$
Find *a* such that 65% of values are *less than a* $P(X < a) = 0.65$	$= NORM.INV(0.65, mean, standard\ deviation)$
Find *a* such that 65% of values are *greater than a* $P(X > a) = 0.65$ Or equivalently $P(X < a) = 0.35$	$= NORM.INV(0.35, mean, standard\ deviation)$
Find *a* and *b* such that 95% of values are between *a* and *b* $P(a < X < b) = 0.95$ Then: $P(X < a) = 0.025$ $P(X < b) = 0.975$	Finding *a*: $= NORM.INV(0.025, mean, standard\ deviation)$ Finding *b*: $= NORM.INV(0.975, mean, standard\ deviation)$

CRITICAL VALUES (CH 8-10)

	Mean (σ known) and proportion	**Mean (σ unknown)**
Lower tail	$= NORM.S.INV(\alpha)$	$= T.INV(\alpha, n-1)$
Two tail	$= NORM.S.INV(\alpha/2)$	$= T.INV(\alpha/2, n-1)$ OR $= T.INV.2T(\alpha, n-1)$
Upper tail	$= NORM.S.INV(1-\alpha)$	$= T.INV(1-\alpha, n-1)$

P-VALUES (CH 9-10)

	Mean (σ known) and proportion	**Mean (σ unknown)**
Lower tail	$= NORM.S.DIST(test\ statistic, TRUE)$	$= T.DIST(test\ statistic, n-1, TRUE)$
Two tail	$= 2 * NORM.S.DIST(-ABS(test\ statistic), TRUE)$	$= 2 * T.DIST(-ABS(test\ statistic), n-1, TRUE)$ OR $T.DIST.2T(test\ statistic, n-l)$
Upper tail	$= 1 - NORM.S.DIST(test\ statistic, TRUE)$	$= 1 - T.DIST(test\ statistic, n-1, TRUE)$ OR $= T.DIST.RT(test\ statistic, n-1)$

EXTRA

Sample mean	$= AVERAGE(select\ cell\ range)$
Sample standard deviation	$= STDEV.S(select\ cell\ range)$
Sample size	$= COUNT(select\ cell\ range\ )$ OR $= COUNTA(select\ cell\ range)$
Standardize value	$= STANDARDIZE(value, mean, standard\ deviation)$